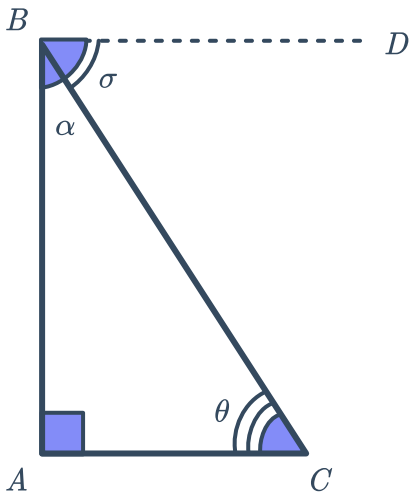
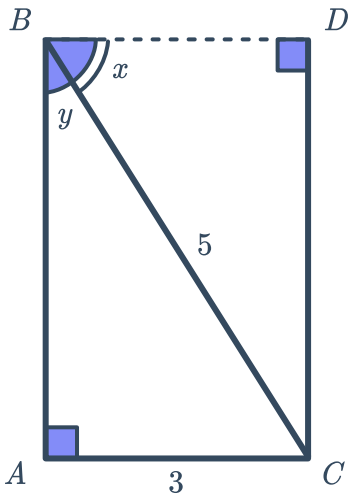
Angles of elevation and depression

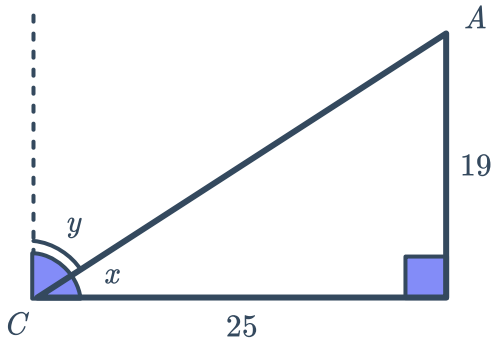
1 A man standing at the top of the tower at point B, is looking at the ground at point C. Identify the angle of depression in the following figure:



2 Find the angle of depression from point B to point C in the diagram shown. Round your answer to two decimal places.

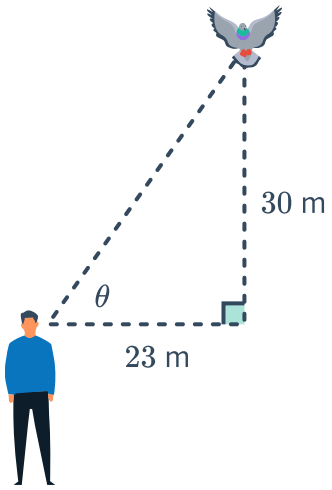


3 Find x, the angle of elevation to point A from point C. Round your answer to two decimal places.

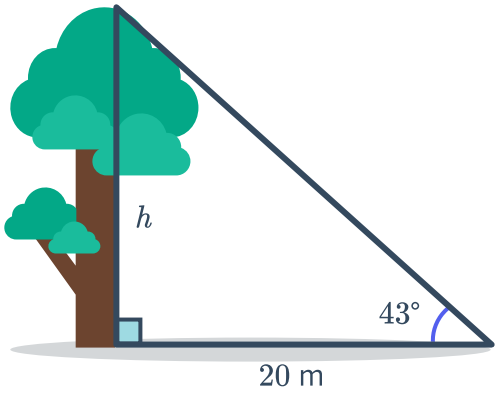


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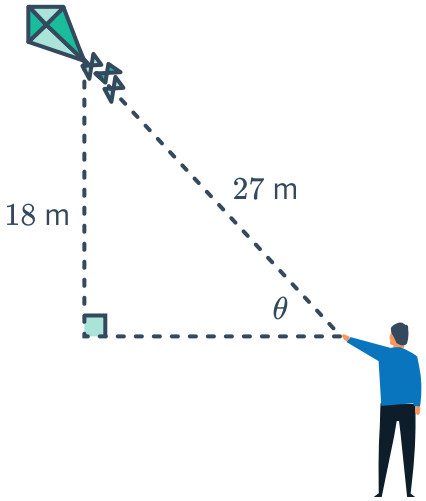
4 The person in the picture sights a pigeon above him. Find the angle of elevation, θ, to two decimal places.



5 Sally measures the angle of elevation to the top of a tree from a point $20\ \text{m}$ away to be 43°. Find the height of the tree, h, to the nearest whole number.

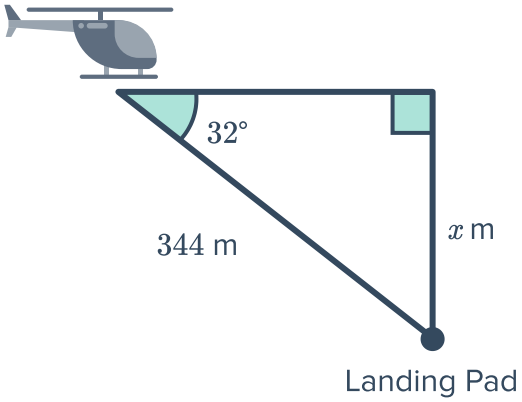


6 A boy flying his kite releases the entire length of his string which measures $27\ \text{m}$, so that the kite is $18\ \text{m}$ above him.
If the angle the string makes with the horizontal ground is θ, find θ to two decimal places.

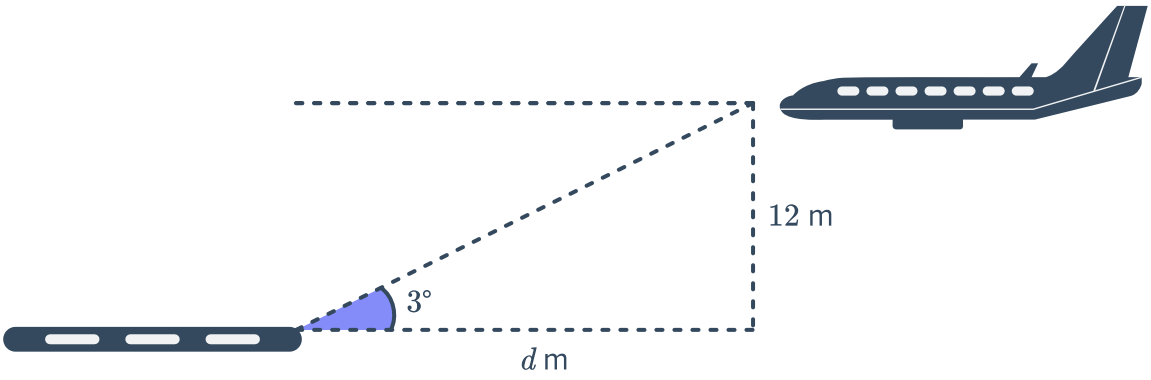


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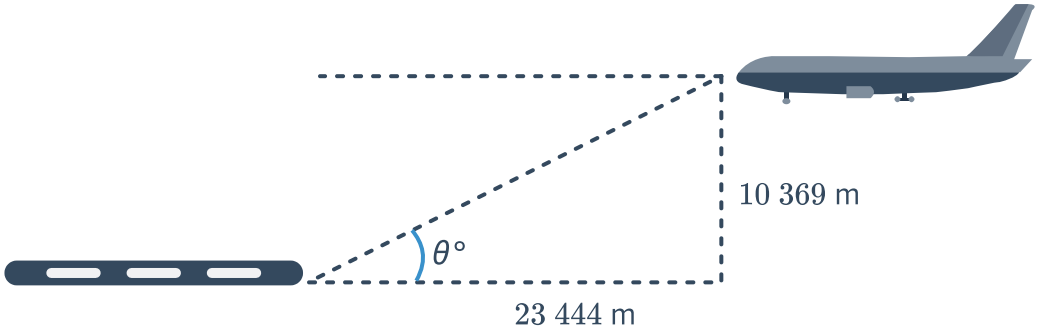
7 A helicopter is $344\ \text{m}$ away from its landing pad. If the angle of depression to the landing pad is 32°, find x, the height of the helicopter above the ground, to the nearest metre.



8 The final approach of an aeroplane when landing requires the pilot to adjust the angle of descent to about 3° as shown in the diagram below. If the plane is $12\ \text{m}$ above the runway and has $d\ \text{m}$ until touchdown, find d to the nearest metre.



9 The airtraffic controller is communicating with a plane in flight approaching an airport for landing. The plane is $10\,369\ \text{m}$ above the ground and is still $23\,444\ \text{m}$ from the runway. If θ° is the angle at which the plane should approach, find θ to one decimal place.



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Applications

10 Fred is on a ship and observes a lighthouse on a cliff in the distance. The base of the cliff is $651\,\text{m}$ away from the ship, and the angle of elevation of the top of the lighthouse from Fred is 35°.

a If the top of the lighthouse is $x\,\text{m}$ above sea level, find x correct to two decimal places.

b If the lighthouse is 30 metres tall, how tall is the cliff that the lighthouse stands on? Round your answer correct to two decimal places.

11 Buzz is standing $250\,\text{m}$ from a building and measures the angle of elevation of the top of the building to be 27°.

a If the difference in height between the top of the building and Buzz's eye is $h\,\text{m}$, find h correct to two decimal places.

b If Buzz's eye is $172\,\text{cm}$ from the ground, find the height of the building correct to one decimal place.

12 From the top of a rocky ledge that is $274\,\text{m}$ high, the angle of depression to a boat is 15°. If the boat is $d\,\text{m}$ from the foot of the cliff, find d correct to two decimal places.

13 Amelia measures the angle of elevation to the top of a tree from a point, $29\,\text{m}$ away from the base, to be 31°. Find the height of the tree, h, to the nearest metre.

14 The angle of elevation from an observer to the top of a tree is 18°. If the distance between the tree and the observer is $d\,\text{m}$ and the tree is known to be $3.53\,\text{m}$ high, find the value of d to two decimal places.

15 At a certain time of the day a light post, $6\,\text{m}$ tall, has a shadow of $5.8\,\text{m}$. If the angle of elevation of the sun at that time is θ°, find θ to two decimal places.

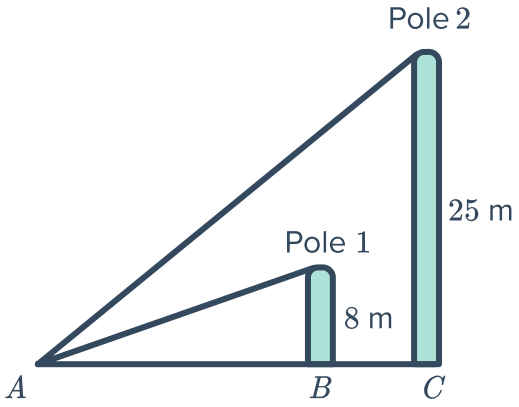
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16 A man stands at point A looking at the top of two poles. Pole 1 has a height $8\ \text{m}$ and an angle of elevation of 34° from point A. Pole 2 has a height $25\ \text{m}$ and an angle of elevation of 57° from point A.

a Find the distance from A to B, to two decimal places.

b Find the distance from A to C, to two decimal places.

c Hence, find BC, the distance between the two poles in metres. Round your answer to one decimal place.

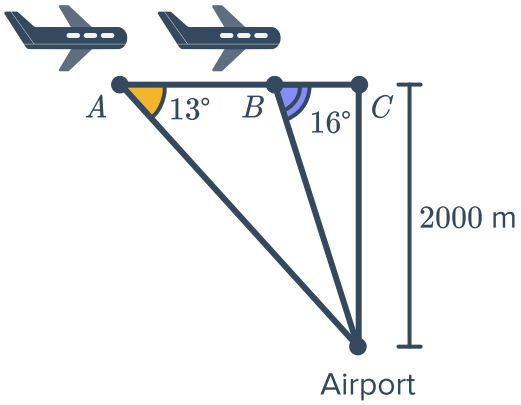


17 A fighter jet, flying at an altitude of $2000\ \text{m}$ is approaching an airport. The pilot measures the angle of depression to the airport to be 13°. One minute later, the pilot measures the angle of depression again and finds it to be 16°.

a Find the distance AC, to the nearest metre.

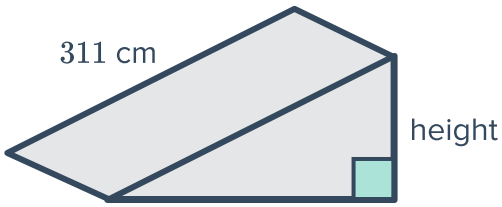
b Find the distance BC, to the nearest metre.

c Hence, find the distance covered by the jet in that one minute, to the nearest metre.



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18 A ramp of length $311\ \text{cm}$ needs to ascend at an angle between 10° and 20° for it to be safe to use.



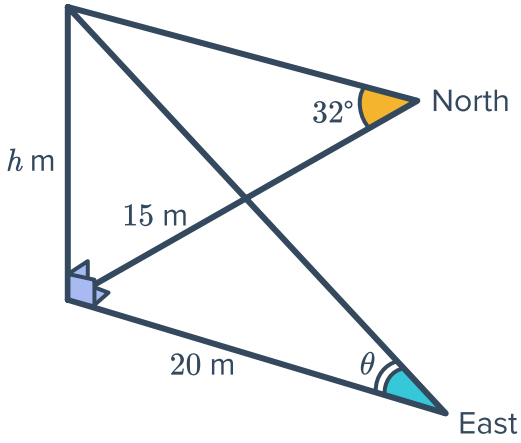
a If the height of the ramp is $152\ \text{cm}$, and the angle the ramp makes with the ground is x, find x to two decimal places.

b If the height of the ramp is $25\ \text{cm}$, and the angle the ramp makes with the ground is y, find y to two decimal places.

c If the height of the ramp is $100\ \text{cm}$, and the angle the ramp makes with the ground is z, find z to two decimal places.

d Hence, at which height is the ramp safe?

19 From a point $15\ \text{m}$ due north of a tower, the angle of elevation of the tower is 32°.



a Find the height of the tower h. Round your answer to two decimals places.

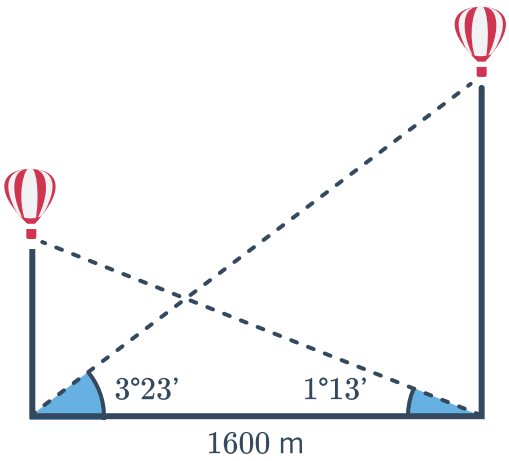
b Find the size θ of the angle of elevation of the tower at a point $20\ \text{m}$ due east of the tower. Round your answer to the nearest degree.

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20 Two hot air balloons are tied to the ground below, each at a different location. An observer at each location measures the angle of elevation to the opposite balloon. The observers are $1600\ \text{m}$ apart as shown in the diagram:

a Calculate the difference in height between the two balloons. Round your answer to the nearest metre.

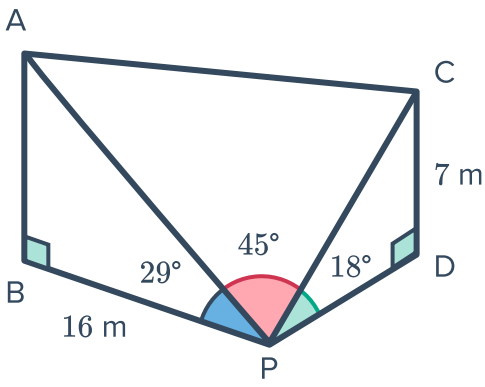
b Hence, find the direct distance between the two balloons. Round your answer to the nearest metre.



21 Roald is standing at point P and observes two poles, AB and CD, of different heights. P, B, and D are on horizontal ground: From P, the angles of elevation to the top of the poles at A and C are 29° and 18° respectively. Roald is $16\ \text{m}$ from the base of pole AB. The height of pole CD is $7\ \text{m}$.

a Calculate the distance from Roald to the top of pole CD, to two decimal places.

b Calculate the distance from Roald to the top of pole AB, to two decimal places.



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